

Glue practical lab

The purpose of the assignment was to You have completed an **AWS Glue ETL pipeline**. The main purpose was to take employee data stored as **JSON**, clean and transform it, convert it to **CSV**, and save the result in a different S3 bucket. This matches the three requirements in your lab.

The final architecture is:

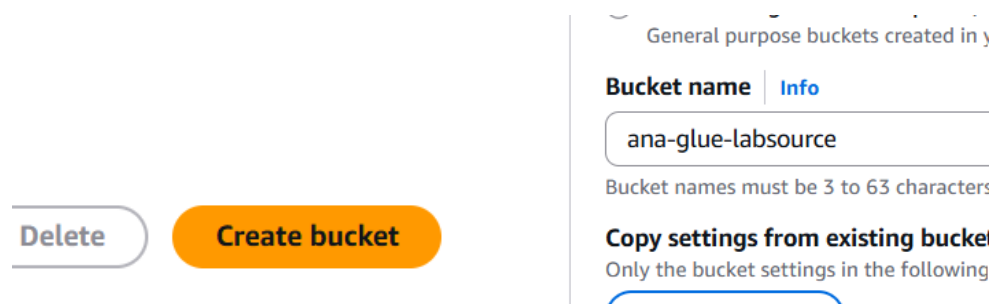
Source S3 bucket → Drop Duplicates → Select Fields → Target S3 bucket (CSV)

1. I created a source S3 bucket

First, I created a new bucket:

ana-glue-labsource

which I used it as the source bucket, where the I uploaded the JSON file into. It was containing duplicates employees, because by the end, the duplicates values had to be removed.



The image shows a screenshot of the AWS S3 console. On the left, there are two buttons: a grey 'Delete' button and an orange 'Create bucket' button. On the right, a panel shows the details for a bucket named 'ana-glue-labsource'. The panel has a 'General purpose buckets created in y' header, a 'Bucket name' field with the value 'ana-glue-labsource', and an 'Info' link. Below the name, it says 'Bucket names must be 3 to 63 characters'. There is also a section for 'Copy settings from existing bucket' with a note 'Only the bucket settings in the following' and a partially visible dropdown menu.

Upload succeeded
For more information, see the [Files and folders](#) table.

After you navigate away from this page, the following information i

Summary

Destination
[s3://ana-glue-labsource](#)

Files and folders | **Configuration**

Files and folders (1 total, 1.4 KB)

Find by name

Name	Folder
emp_json,json,json	-

- For this assignment, a second bucket was needed, as the target bucket, where it's purpose was to receive the processed CSV file from glue.

ana-glue-labtarget

General purpose buckets created in your a

Bucket name | [Info](#)

[ana-glue-labtarget](#)

Bucket names must be 3 to 63 characters and i

- Next step is to go to AWS glue -> ETL jobs -> Visual ETL

Visual ETL let's us build and ETL pipeline using boxes, nodes, instead of writing all the Spark code ourselves. E -> extract - read the data, T -> transform – change or clean the data, L – load – save it somewhere

AWS Glue

Getting started

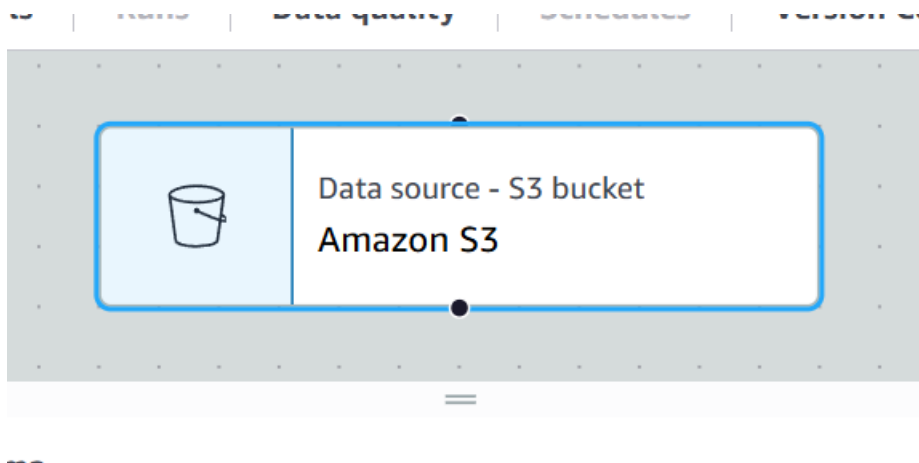
SageMaker Unified Studio [↗](#)

ETL jobs

Visual ETL

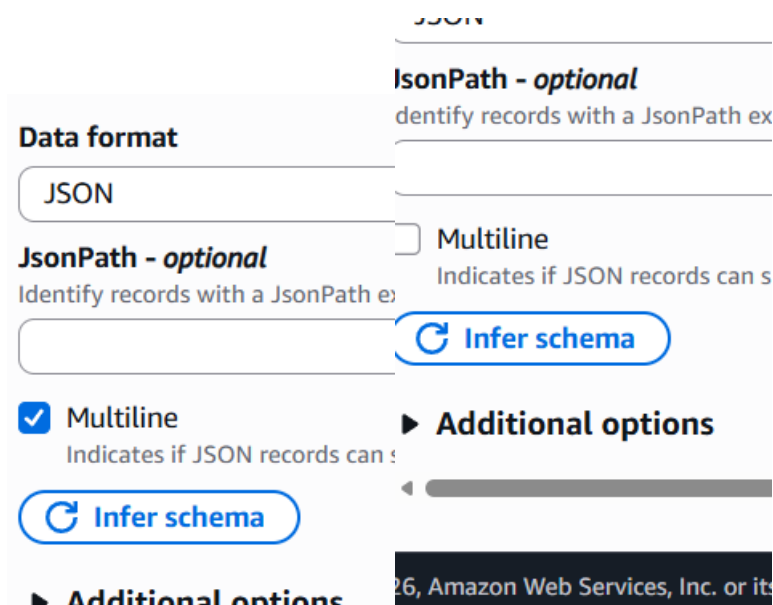
Notebooks

4. I've added Amazon S3 as the data source, which represents the extract part, Glue then reads the file so it can process the record.

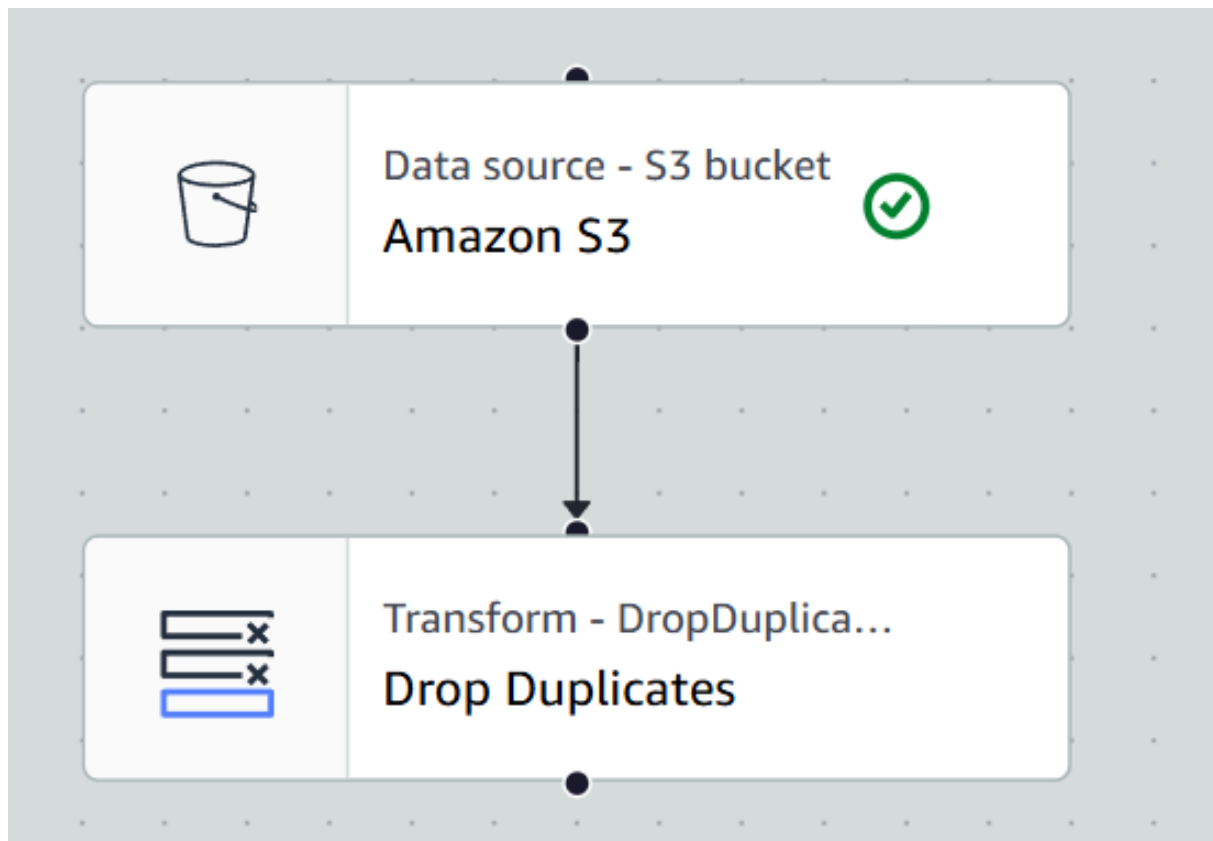


5. The next step was to tell Glue that the Data format is Json. It is important to let Glue know which type of data format we have, because it must know how to interpret the file. Initially I had Multiline selected and I removed it, because Multiline tells Glue that a Json record can extend over several lines. I removed it because for the csv file that we had, multiline was not necessarily.

I used infer schema which describe the structure of the data, Glue examined the data and now it knows both the column names and their data type.



6. I added Drop Duplicates to remove repeated records. I selected match specific keys -> emp-id, which means that Glue is using emp_id to determine if two records represent duplicates.



```
1  [
2    {
3      "emp_id": 1,
4      "name": "manish",
5      "salary": 10000,
6      "address": "india",
7      "loc": "",
8      "email": ""
9    },
10   {
```

Drop duplicates | Info

Remove rows of repeating data from y

☐ Match entire rows (Distinct)
Use all fields to compare between them.

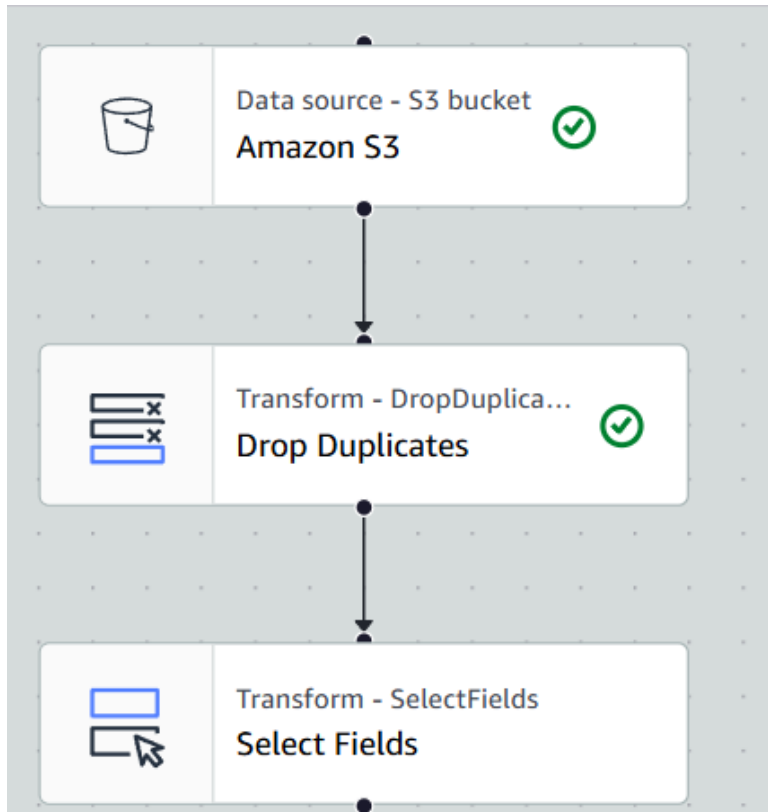
☒ Match specific keys
Choose what fields will be used to

Keys to match rows
Choose fields for matching duplicate r

Choose one or more field

emp_id X

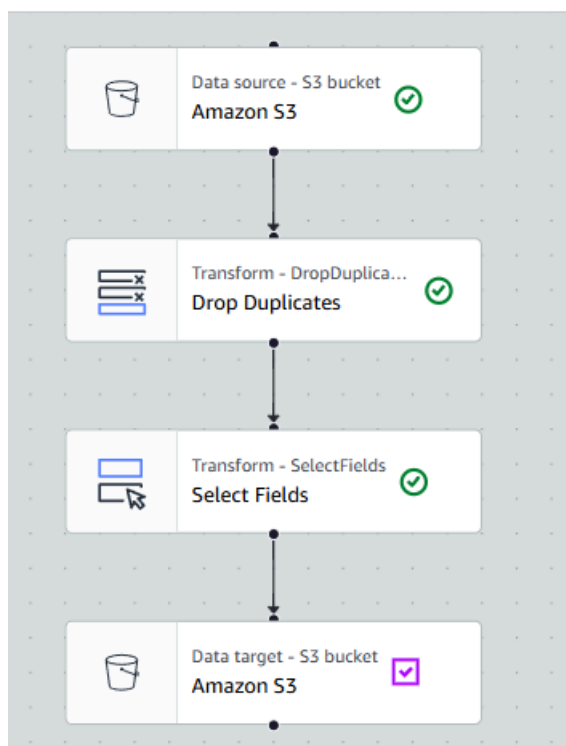
7. I added Select Fields and I chose 4 fields emp_id, name, salary, address. So I am telling Glue that I do not need loc and email, so keep only this 4 fields.



SelectFields

☐	Field	Data type
☒	emp_id	int
☒	name	string
☒	salary	int
☒	address	string
☐	loc	string
☐	email	string

8. I've added a second Amazon S3 node and with this the pipeline is complete. The second S3 node, will receive and transform the data, the last node represents the Load part of the ETL. In the Node parents I have selected "select fields", because the target comes immediately after select fields, which ensures that the target receives the cleanest and selected data, rather than the original source data.



9. Firstly I tried the file format with CSV and compression type Snappy, by mistake, which is totally fine, because, because Glue can transform and structure/content and also change the file format and after I change it to JSON also the request number of rows was set to 10, so I have changed that one as well, to 2. Also Snappy compression makes files smaller, so instead of writing an uncompressed file, and after I changed to none.

Node parents
Choose which nodes will provide inputs for this one.

Choose one or more parent node

Select Fields
SelectFields - Transform

Format
CSV

Compression Type
Snappy

S3 Target Location
Choose an S3 location in the format s3://bucket/prefix/object/ with a trailing slash

Q s3://ana-glue-labtarget/ X

View ↗

Brow

Data Catalog update options | Info

Choose which nodes will provide inputs for this one.

Choose one or more parent node

Select Fields
SelectFields - Transform

Format
JSON

Compression Type
None

Description - optional

Descriptions can be up to 2048 characters long.

IAM Role

Role assumed by the job with permission to access your data stores. Ensure that this role has permission to you targets, temporary directory, scripts, and any libraries used by the job.

AWSGlueServiceRole-Ana-salesproject ▼

↻

Type

The type of ETL job. This is set automatically based on the types of data sources you have selected.

Spark

Glue version | Info

10. After running the job, we can see that it was successful and the target bucket, received the output from the input bucket.

(4vCPU and 16GB RAM)

Automatically scale the number of workers

☐ AWS Glue will optimize costs and resource u throughout the job run. Requires Glue 3.0 or

Requested number of workers

The number of workers you want AWS Glue to allocate t

2

Filter job runs by property

	Run status	Retries	Start time (Local)	End time (Local)	Duration
☐	Succeeded	0	08/11/2026 11:33:03	08/11/2026 11:34:25	1 m 14 s
☒	Succeeded	0	08/11/2026 11:27:18	08/11/2026 11:28:47	1 m 21 s

permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	00028.snappy	snappy	(UTC+01:00)		
☐	run-1786444103231-part-r-00029.snappy	snappy	August 11, 2026, 11:28:37 (UTC+01:00)	4.0 B	Standard
☐	run-1786444103231-part-r-00030.snappy	snappy	August 11, 2026, 11:28:31 (UTC+01:00)	67.0 B	Standard
☐	run-1786444103231-part-r-00031.snappy	snappy	August 11, 2026, 11:28:37 (UTC+01:00)	4.0 B	Standard
☐	run-1786444103231-part-r-00032.snappy	snappy	August 11, 2026, 11:28:31 (UTC+01:00)	4.0 B	Standard
☐	run-1786444103231-part-r-00033.snappy	snappy	August 11, 2026, 11:28:31 (UTC+01:00)	4.0 B	Standard
☐	run-1786444103231-part-r-00034.snappy	snappy	August 11, 2026, 11:28:31 (UTC+01:00)	4.0 B	Standard
☐	run-1786444103231-part-r-00035.snappy	snappy	August 11, 2026, 11:28:31 (UTC+01:00)	4.0 B	Standard
☐	run-1786444452766-part-r-00000	-	August 11, 2026, 11:34:14 (UTC+01:00)	349.0 B	Standard

Conclusion

In this practical, I learned how to use **AWS Glue and Amazon S3** to work with data. I created a Glue job that reads employee data from an S3 bucket, removes duplicate records, and keeps only the fields that I need.

I also learned that **IAM roles give AWS Glue permission** to access the S3 buckets. After setting up the job, I ran it successfully and checked the results in my target S3 bucket.

Overall, this practical helped me understand the basic idea of **ETL – Extract, Transform and Load**, and how AWS Glue can be used to clean and process data automatically.